
KTU
NOTES

Closed loop poles are the roots of the characteristic equation, $1+G(s)H(s)=$

$$F(s)=1+G(s)H(s)=\frac{N_1(s)}{D_1(s)}$$

Poles of closed loop transfer function are given by $F(s)=0$ or $N_1(s)=0$.

Hence, poles of the closed-loop transfer function are the zeros of q(s).

1	Poles of $F(s)=1+G(s)H(s)$ are the open loop poles
2	Zeros of $F(s)=1+G(s)H(s)$ are the closed loop poles
3	For a stable system, the zeros of F(s) should not lie in the RHP.

Given $G(s)=\frac{2(s+1)}{(s-1)}$. Is the open loop system stable? Is the closed loop sys

$$G(s)=\frac{2(s+1)}{(s-1)}$$

Open loop system has pole in the right-hand side of s-plane. Hence it is not stab

$$\frac{C(s)}{R(s)}=\frac{2(s+1)}{s-1+2(s+1)}=\frac{\frac{2}{3}(s+1)}{s+\frac{1}{3}}$$

Closed system is stable because only pole is at s = -1/3.

Given $G(s)=\frac{2(s-1)}{(s+1)}$. Is the open loop system stable? Is the closed loop sys

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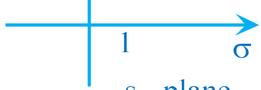
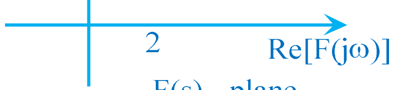
$$\frac{C(s)}{R(s)}=\frac{2(s-1)}{s+1+2(s-1)}=\frac{\frac{2}{3}(s-1)}{s-\frac{1}{3}}$$



s - plane



F(s) - plane

Closed contour: - A closed contour in a complex plane is a continuous curve b
the same point.

Enclosed: - A point or region is said to be enclosed by a closed path if the po
right of the path when the path is traversed in the prescribed direction.

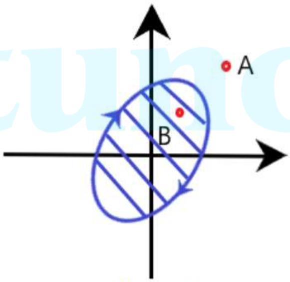


Figure 1

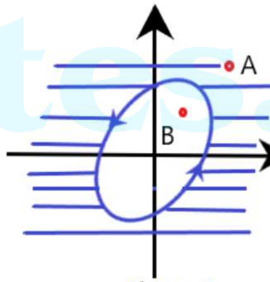
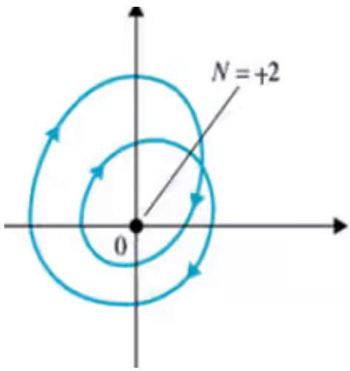


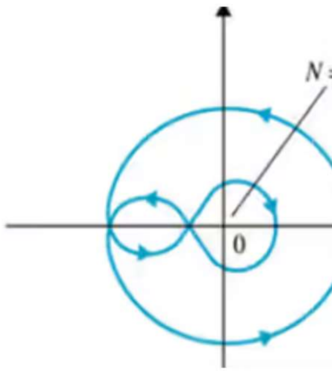
Figure 2

In figure (1), point B is enclosed; but A is not enclosed. In figure (2), point A i
enclosed.

Encircled: - A point is said to be encircled by a closed path if it lies inside that
figure (1) and (2) are encircled.

No. of encirclements N: -





arbitrary closed path C is chosen in the s-plane so that the path does not go thr
or zeros of F(s), the corresponding C_s locus mapped in the F(s) plane will encir
times as the difference between the number of zeros and poles of F(s) that a
plane locus C.

In equation form, the principle of argument is stated as

$N = Z - P$

where

N = number of encirclements of the origin made by the F(s) plane locus C_s

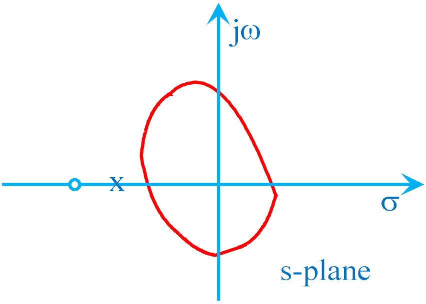
Z = number of zeros of F(s) encircled by the s-plane locus C in the s plane

P = number of poles of F(s) encircled by the s plane locus C in the s plane

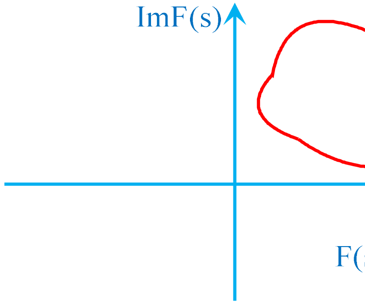
For $Z = 4, P = 2, N = 2 \rightarrow$ 2 clockwise encirclements

$Z = 2, P = 2, N = 0 \rightarrow$ No encirclements

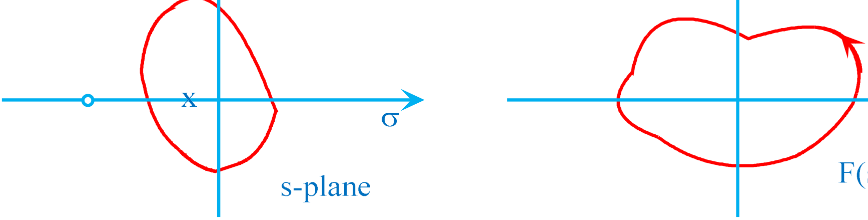
$Z = 2, P = 4, N = -2 \rightarrow$ 2 counter-clockwise encirclements



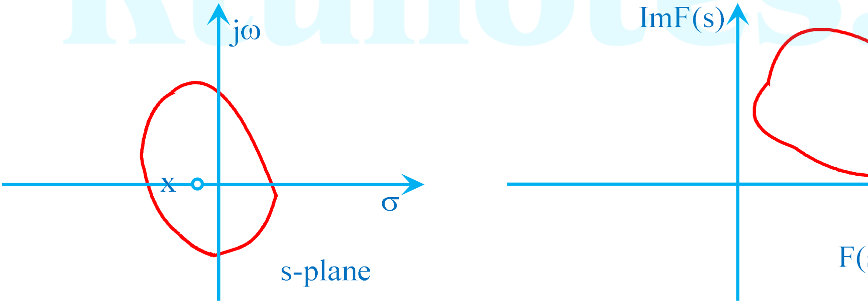
No poles or zeros are covered



No. of encirclement



One pole covered No. of encirclements of origin = -1



One zero & one pole covered No. of encirclements o

Note: Zeros of $1+G(s)H(s)$ are the poles of the closed-loop transfer function

Nyquist suggested that rather than analyzing whether all the zeros are located plane, it is better to examine the presence of any one zero of $1+G(s)H(s)$ in plane making the system unstable.

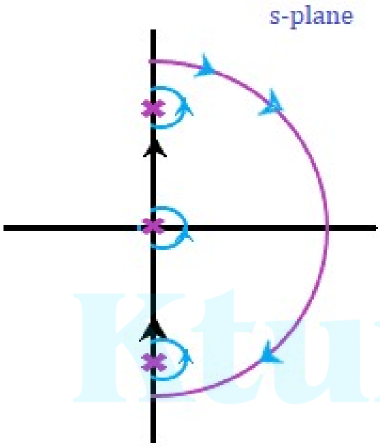
Since $G(s)H(s)=-1$, it is convenient to choose the $G(s)H(s)$ plane instead of F(s) plane for investigating the stability. Hence, the relation N = Z – P can still be used for finding the unstable zeros of F(s) where N is the encirclement of the point (-1 + j0) in the GH plane by the Nyquist path C_{GH}.

SELECTION OF NYQUIST CONTOUR

The Nyquist contour is selected such that it encloses the entire right-half side o the s-plane. It should enclose all the poles and zeroes present on the right-half o singular points. The singular points are considered as the points lying on the these points are avoided.

If the transfer function has a pole at origin, the contour encloses all the poles origin.

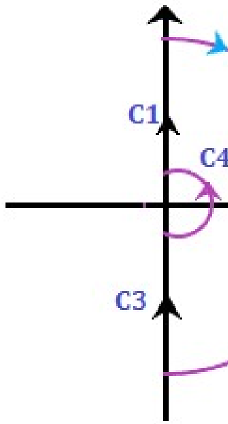
If there are no poles on the imaginary axis, the contour will appear as:



MAPPING A CONTOUR

A singular point (point on the imaginary axis) is not analytical. Hence, it is generally avoided. The Nyquist contour is mapped to determine the encirclement of the point -1 + j0. The contour is drawn based on the transfer function G(s)H(s).

Generally, there are four sections C_1, C_2, C_3, and C_4. The Nyquist criterion plot is divided into four sections so that the mapping can be easily carried out section wise. At last, all the sections are combined to produce the desired Nyquist plot. Let the four sections be:



Mapping of section C_1

The value of ω in section C_1 ranges from 0 to +∞. The contour will be drawn respect to the above range, and it will be the locus plot of G(jω)H(jω). For mappi the values of G(jω)H(jω) for various values of ω and sketch the actual locus of

Mapping of section C_2

NYQUIST STABILITY CRITERION

If the contour C_{GH} of the open-loop transfer function G(s)H(s) in the G(s)H(s
to the Nyquist contour in the s-plane encircles the point -1+j0 in the counter c
many as time as the right-half s-plane poles of G(s)H(s), then the closed-loop

If P = number of poles of G(s)H(s) or F(s) in the right-half of the s-plane
N = number of clockwise encirclements of (-1+j0) point in the GH plan
Z = Number of zeros of F(s) = 1 + G(s)H(s) in the right half of the s-pla

To make a closed-loop system stable, there must not be any zero in the right-hal
Z = 0.

$$Z = P + N = 0$$

If $Z = P + N \neq 0$, closed loop system is unstable.

Note: Positive sign for clockwise encirclement and negative sign for anticlockw

Sketch the Nyquist plot for the open loop transfer function $G(s)H(s) = \frac{\quad}{(s}$
determine the stability of the closed loop system by Nyquist criterion.

$$G(s)H(s) = \frac{25}{(1+s)(1+0.5s)}$$

Open loop poles in right side of s–plane = P = 0

$$G(j\omega)H(j\omega) = \frac{25}{(1+j\omega)(1+j0.5\omega)}$$

$$\left|G(j\omega)H(j\omega)\right| = \frac{25}{\sqrt{1+\omega^2} \times \sqrt{1+(0.5\omega)^2}}$$

	$\lvert G(j\omega)H(j\omega)\rvert$	
0	25	0
∞	0	-18

MAPPING OF C_2

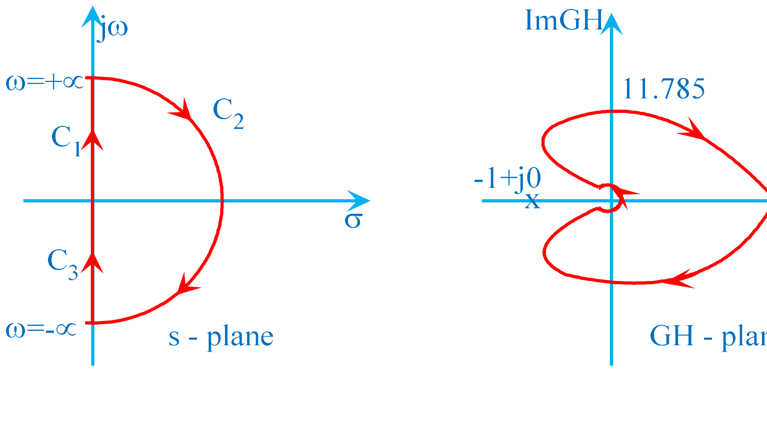
$$s = \underset{R\to\infty}{Lt}\ \mathrm{R}e^{j\theta}\ ;\ \theta = +\frac{\pi}{2}\ \text{ to }\ -\frac{\pi}{2}\ \text{clockwise}$$

$$G(s)H(s)\Big|_{s=\mathrm{Re}^{j\theta}} = \frac{50}{(\mathrm{Re}^{j\theta}+1)(\mathrm{Re}^{j\theta}+2)} \approx \frac{50}{\mathrm{Re}^{j\theta}\times \mathrm{Re}^{j\theta}} = \frac{50}{\mathrm{R}^2 e^{j2\theta}}$$

$$\underset{R\to\infty}{Lt}\ GH(\mathrm{Re}^{j\theta}) = \underset{\varepsilon\to 0}{Lt}\ \varepsilon\, e^{-j2\theta}$$

θ	$\underset{R\to\infty}{Lt}\ GH(\mathrm{Re}^{j\theta})$
$+\frac{\pi}{2}$	$\underset{\varepsilon\to 0}{Lt}\ \varepsilon\, e^{-j\pi}$
$-\frac{\pi}{2}$	$\underset{\varepsilon\to 0}{Lt}\ \varepsilon\, e^{+j\pi}$

C_2 in s-plane is mapped into a circular arc of radius 0 at origin in the GH-plane.



$$G(s)H(s) = \frac{s+2}{(s+1)(s-1)} = \frac{2(1+0.5s)}{(1+s)(-1+s)}$$

Open loop poles in right side of s–plane = P = 1

$$G(j\omega)H(j\omega) = \frac{2(1+j0.5\omega)}{(1+j\omega)(-1+j\omega)}$$

$$\left|G(j\omega)H(j\omega)\right| = \frac{2\sqrt{1+(0.5\omega)^2}}{\sqrt{1+\omega^2}\times\sqrt{1+\omega^2}} = \frac{2\sqrt{1+(0.5\omega)^2}}{1+\omega^2}$$

$$\angle G(j\omega)H(j\omega) = \tan^{-1}0.5\omega - \tan^{-1}\omega + \tan^{-1}\omega - 180 = \tan^{-1}0.5\omega - 180°$$

MAPPING OF C_1 & C_3

ω	$\lvert G(j\omega)H(j\omega)\rvert$	$\angle G(j\omega)$
0	2	-180
∞	0	-90

MAPPING OF C_2

$$s = \underset{R\to\infty}{Lt}\ \mathrm{R}e^{j\theta};\ \theta = +\frac{\pi}{2}\ \text{ to }\ -\frac{\pi}{2}\ \text{ clockwise}$$

$$G(s)H(s)\Big|_{s=\mathrm{R}e^{j\theta}} = \frac{\mathrm{R}e^{j\theta}+2}{(\mathrm{R}e^{j\theta}+1)(\mathrm{R}e^{j\theta}-1)} \approx \frac{\mathrm{R}e^{j\theta}}{\mathrm{R}e^{j\theta}\times\mathrm{R}e^{j\theta}} = \frac{1}{\mathrm{R}e^{j\theta}}$$

$$\underset{R\to\infty}{Lt}\ GH(\mathrm{R}e^{j\theta}) = \underset{\varepsilon\to 0}{Lt}\ \varepsilon\, e^{-j\theta}$$

θ	$\underset{R\to\infty}{Lt}\ GH(\mathrm{R}e^{j\theta})$
$+\frac{\pi}{2}$	$\underset{\varepsilon\to 0}{Lt}\ \varepsilon\, e^{-j\pi/2}$
$-\frac{\pi}{2}$	$\underset{\varepsilon\to 0}{Lt}\ \varepsilon\, e^{+j\pi/2}$

C_2 in s-plane is mapped into a circular arc of radius 0 at origin in the GH-plane.

Z = P + N = 0; system is stable.

Sketch the Nyquist plot for the open-loop transfer function $G(s)H(s) =$

determine the stability of the closed loop system by Nyquist criterion.

$$G(s)H(s) = \frac{50}{s(s+5)} = \frac{10}{s(1+0.2s)}$$

Open loop poles in right side of s–plane = P = 0

$$G(j\omega)H(j\omega) = \frac{10}{j\omega(1+j0.2\omega)}$$

$$\left|G(j\omega)H(j\omega)\right| = \frac{10}{\omega \times \sqrt{1+(0.2\omega)^2}}$$

$$\angle G(j\omega)H(j\omega) = -90 - \tan^{-1} 0.2\omega$$

MAPPING OF C_1 & C_3

ω	$\left\|G(j\omega)H(j\omega)\right\|$	$\angle G(j\omega)H$
0	∞	-90
∞	0	-180

MAPPING OF C_2

$$s = \underset{R\to\infty}{Lt}\ \mathrm{R}e^{j\theta};\ \theta = +\frac{\pi}{2} \text{ to } -\frac{\pi}{2} \text{ clockwise}$$

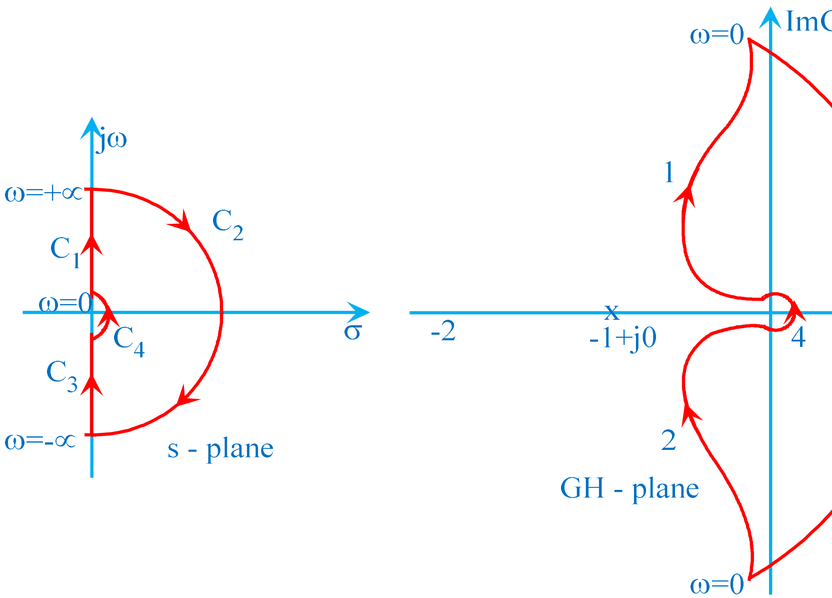
$$G(s)H(s)\Big|_{s=\mathrm{R}e^{j\theta}} = \frac{50}{\mathrm{R}e^{j\theta}(\mathrm{R}e^{j\theta}+5)} \approx \frac{50}{\mathrm{R}e^{j\theta}\times\mathrm{R}e^{j\theta}} = \frac{1}{\mathrm{R}^2 e^{j2\theta}}$$

$$\underset{R\to\infty}{Lt}\ GH(\mathrm{R}e^{j\theta}) = \underset{\varepsilon\to 0}{Lt}\ \varepsilon\, e^{-j2\theta}$$

$$\underset{\varepsilon \to 0}{Lt}\, GH(\varepsilon\, e^{j\theta}) = \propto e^{-j\theta}$$

θ	$\underset{\varepsilon \to 0}{Lt}\, GH(\varepsilon\, e^{j\theta})$
$-\dfrac{\pi}{2}$	$\propto e^{j\pi/2}$
$+\dfrac{\pi}{2}$	$\propto e^{-j\pi/2}$

C_4 in s-plane is mapped into a semicircle of infinite radius with angle varying fr



Number of encirclements of point -1+j0, N = 0.

$$\left|G(j\omega)H(j\omega)\right| = \frac{12.5}{\omega \times \sqrt{1+(0.25\omega)^2} \times \sqrt{1+\omega^2}}$$

$$\angle G(j\omega)H(j\omega) = -90 - \tan^{-1} 0.25\omega + \tan^{-1}\omega - 180 = -270 - \tan^{-1} 0.25\omega + t$$

MAPPING OF C_1 & C_3

ω	$\left\|G(j\omega)H(j\omega)\right\|$	$\angle G(j\omega)$
0	∞	-270
∞	0	-270

MAPPING OF C_2

$$s = \underset{R\to\infty}{Lt}\ \mathrm{R}e^{j\theta}\ ;\ \theta = +\frac{\pi}{2}\ \text{to}\ -\frac{\pi}{2}\ \text{clockwise}$$

$$G(s)H(s)\Big|_{s=\mathrm{R}e^{j\theta}} = \frac{50}{\mathrm{R}e^{j\theta}(\mathrm{R}e^{j\theta}+4)(\mathrm{R}e^{j\theta}-1)} \approx \frac{50}{\mathrm{R}e^{j\theta}\times \mathrm{R}e^{j\theta}\times \mathrm{R}e^{j\theta}} = \frac{1}{\mathrm{R}^3 e^{j3\theta}}$$

$$\underset{R\to\infty}{Lt}\ GH(\mathrm{R}e^{j\theta}) = \underset{\varepsilon\to 0}{Lt}\ \varepsilon\, e^{-j3\theta}$$

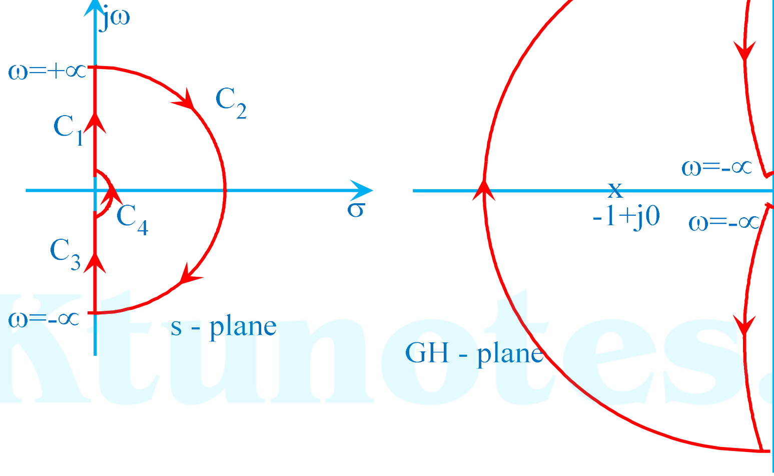
θ	$\underset{R\to\infty}{Lt}\ GH(\mathrm{R}e^{j\theta})$
$+\frac{\pi}{2}$	$\underset{\varepsilon\to 0}{Lt}\ \varepsilon\, e^{-j3\pi/2} = \underset{\varepsilon\to 0}{Lt}\ \varepsilon\, e^{+j\pi/2}$
$-\frac{\pi}{2}$	$\underset{\varepsilon\to 0}{Lt}\ \varepsilon\, e^{+j3\pi/2} = \underset{\varepsilon\to 0}{Lt}\ \varepsilon\, e^{-j\pi/2}$

C_2 in s-plane is mapped into a circular arc of radius 0 at origin in the GH-plane.

MAPPING OF C_3

$$s = \underset{\varepsilon\to\infty}{Lt}\ \varepsilon\, e^{j\theta}\ ;\ \theta = -\frac{\pi}{2}\ \text{to}\ +\frac{\pi}{2}\ \text{clockwise}$$

$$G(s)H(s)\Big|_{s=\varepsilon e^{j\theta}} = \frac{50}{\varepsilon e^{j\theta}(\varepsilon e^{j\theta}+4)(\varepsilon e^{j\theta}-1)} \approx \frac{50}{-\varepsilon e^{j\theta}\times 4} = \frac{-12.5}{\varepsilon e^{j\theta}} \approx \infty\ e^{-j(180+\theta)}$$



Number of encirclements of point -1+j0, N = 1 (clockwise)

P = 1

Z = P + N = 1 +1 = 2; system is unstable.

Sketch the Nyquist plot for the open-loop transfer function $G(s)H(s) = \frac{}{s(s}$

determine the stability of the closed loop system by Nyquist criterion.

$$G(s)H(s) = \frac{5(s-3)}{s(s+1)(s-1)} = \frac{5/3(-1+0.333s)}{s(1+s)(-1+s)}$$

Open loop poles in right side of s–plane = P = 1

$$G(j\omega)H(j\omega) = \frac{5/3(-1+j0.333\omega)}{j\omega(1+j\omega)(-1+j\omega)}$$

$$\left|G(j\omega)H(j\omega)\right| = \frac{5/3\sqrt{1+(0.333\omega)^2}}{\omega\times\sqrt{1+\omega^2}\times\sqrt{1+\omega^2}}$$

$$\angle G(j\omega)H(j\omega) = 180-\tan^{-1}0.333\omega-90-\tan^{-1}\omega-(180-\tan^{-1}\omega) = -90-$$

MAPPING OF C_1 & C_3

$+\dfrac{\pi}{2}$	$\underset{\varepsilon\to 0}{Lt}\ \varepsilon\, e^{-j\pi}$
$-\dfrac{\pi}{2}$	$\underset{\varepsilon\to 0}{Lt}\ \varepsilon\, e^{+j\pi}$

C_2 in s-plane is mapped into a circular arc of radius 0 at origin in the GH-plane.

MAPPING OF C_4

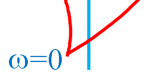
$$s = \underset{\varepsilon\to\infty}{Lt}\ \varepsilon\, e^{j\theta}\ ;\ \theta = -\frac{\pi}{2}\ \text{ to }\ +\frac{\pi}{2}\ \text{ clockwise}$$

$$G(s)H(s)\Big|_{s=\varepsilon e^{j\theta}} = \frac{5(\varepsilon e^{j\theta}-3)}{\varepsilon e^{j\theta}(\varepsilon e^{j\theta}+1)(\varepsilon e^{j\theta}-1)} \approx \frac{5/3}{\varepsilon e^{j\theta}} \approx \infty\, e^{-j\theta}$$

$$\underset{\varepsilon\to 0}{Lt}\ GH(\varepsilon e^{j\theta}) = \infty\, e^{-j\theta}$$

θ	$\underset{\varepsilon\to 0}{Lt}\ GH(\varepsilon e^{j\theta})$
$-\dfrac{\pi}{2}$	$\infty\ e^{+j\pi/2}$
$+\dfrac{\pi}{2}$	$\infty\ e^{-j\pi/2}$

C_4 in s-plane is mapped into a semicircle of infinite radius with angle varying fi



Number of encirclements of point -1+j0, N = 0.

P = 1

Z = P + N = 1; system is unstable.

Sketch the Nyquist plot for the open-loop transfer function $G(s)H(s) = \frac{}{(s}$

and check whether the system is stable for this gain. Find the range of K for th

$$G(s)H(s) = \frac{K(1-s)}{(s+2)(s+3)}$$

Open loop poles in right side of s–plane = P = 0

$$G(j\omega)H(j\omega) = \frac{2(1-j\omega)}{(2+j\omega)(3+j\omega)}$$

$$\left|G(j\omega)H(j\omega)\right| = \frac{2\sqrt{1+\omega^2}}{\sqrt{4+\omega^2}\times\sqrt{9+\omega^2}}$$

$$\angle G(j\omega)H(j\omega) = -\tan^{-1}\omega - \tan^{-1}0.5\omega - \tan^{-1}0.33\omega$$

MAPPING OF C_1 & C_3

ω	$\left\|G(j\omega)H(j\omega)\right\|$	$\angle G(j\omega)$
0	1/3	0°
∞	0	-270

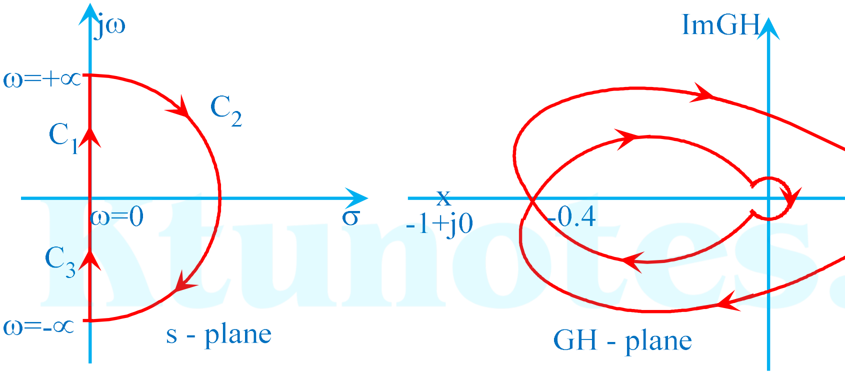
MAPPING OF C_2

$$s = \underset{R\to\infty}{Lt}\ \mathrm{R}e^{j\theta};\ \theta = +\frac{\pi}{2} \text{ to } -\frac{\pi}{2} \text{ clockwise}$$

$$G(s)H(s)\Big|_{s=\mathrm{R}e^{j\theta}} = \frac{2(1-\mathrm{R}e^{j\theta})}{(\mathrm{R}e^{j\theta}+2)(\mathrm{R}e^{j\theta}-3)} \approx \frac{-\mathrm{R}e^{j\theta}}{\mathrm{R}e^{j\theta}\times\mathrm{R}e^{j\theta}} = \frac{-1}{\mathrm{R}e^{j\theta}}$$

$$\underset{R\to\infty}{Lt}\ GH(\mathrm{R}e^{j\theta}) = \underset{\varepsilon\to 0}{Lt}\ \varepsilon\, e^{-j(180-\theta)}$$

$$\left|G(j\omega)H(j\omega)\right| = \frac{\quad}{\sqrt{4+\omega^2}\times\sqrt{9+\omega^2}} = \frac{\quad}{\sqrt{4+3.318^2}\times\sqrt{9+3.318^2}} = 0.4$$



Number of encirclements of point -1+j0, N = 0.

P = 0

Z = P + N = 0; system is stable.

To find the value of K when ω = 3.318 rad/s,

$$\left|G(j\omega)H(j\omega)\right|_{\omega=3.318} = \frac{K\sqrt{1+\omega^2}}{\sqrt{4+\omega^2}\times\sqrt{9+\omega^2}} = \frac{K\sqrt{1+3.318^2}}{\sqrt{4+3.318^2}\times\sqrt{9+3.318^2}} = 0.$$

N = 0 if 0.2K < 1

K < 5

Hence, the system is stable for 0 < K < 5.

> Sketch the Nyquist plot for the open-loop transfer function $G(s)H(s) = \frac{\quad}{s(s}$
>
> determine the stability of the closed loop system by Nyquist criterion.

$$G(s)H(s) = \frac{5}{s(s+1)(s+2)} = \frac{2.5}{s(1+s)(1+0.5s)}$$

$$\angle G(j\omega)H(j\omega) = -90 - \tan^{-1}\omega - \tan^{-1}0.5\omega = -180$$

Hence, $\omega = \sqrt{2} = 1.414$ rad/sec

$$\left|G(j\omega)H(j\omega)\right| = \frac{5}{\omega\sqrt{1+\omega^2}\times\sqrt{4+\omega^2}} = \frac{5}{\sqrt{2}\times\sqrt{1+2}\times\sqrt{4+2}} = 0.833$$

MAPPING OF C_2

$$s = \underset{R\to\infty}{Lt}\ \mathrm{R}e^{j\theta}\ ;\ \theta = +\frac{\pi}{2}\ \text{ to }\ -\frac{\pi}{2}\ \text{ clockwise}$$

$$G(s)H(s)\Big|_{s=\mathrm{R}e^{j\theta}} = \frac{5}{\mathrm{R}e^{j\theta}(\mathrm{R}e^{j\theta}+1)(\mathrm{R}e^{j\theta}+2)} \approx \frac{5}{\mathrm{R}e^{j\theta}\times\mathrm{R}e^{j\theta}\times\mathrm{R}e^{j\theta}} = \frac{1}{\mathrm{R}^3 e^{j3\theta}}$$

$$\underset{R\to\infty}{Lt}\ GH(\mathrm{R}e^{j\theta}) = \underset{\varepsilon\to 0}{Lt}\ \varepsilon\, e^{-j3\theta}$$

θ	$\underset{R\to\infty}{Lt}\ GH(\mathrm{R}e^{j\theta})$
$+\frac{\pi}{2}$	$\underset{\varepsilon\to 0}{Lt}\ \varepsilon\, e^{-j3\pi/2} = \underset{\varepsilon\to 0}{Lt}\ \varepsilon\, e^{+j\pi/2}$
$-\frac{\pi}{2}$	$\underset{\varepsilon\to 0}{Lt}\ \varepsilon\, e^{+j3\pi/2} = \underset{\varepsilon\to 0}{Lt}\ \varepsilon\, e^{-j\pi/2}$

C_2 in s-plane is mapped into a circular arc of radius 0 at origin in the GH-plane.

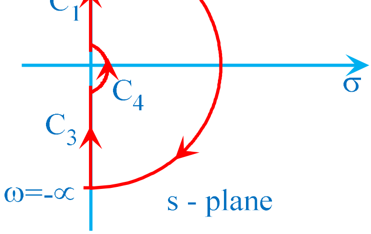
MAPPING OF C_3

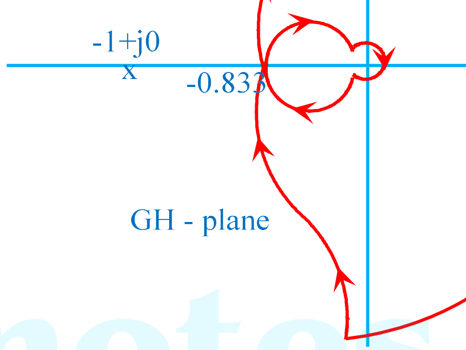
$$s = \underset{\varepsilon\to\infty}{Lt}\ \varepsilon\, e^{j\theta}\ ;\ \theta = -\frac{\pi}{2}\ \text{ to }\ +\frac{\pi}{2}\ \text{ clockwise}$$

$$G(s)H(s)\Big|_{s=\varepsilon e^{j\theta}} = \frac{5}{\varepsilon e^{j\theta}(\varepsilon e^{j\theta}+1)(\varepsilon e^{j\theta}+2)} \approx \frac{5}{\varepsilon e^{j\theta}\times 2} = \frac{2.5}{\varepsilon e^{j\theta}} \approx \infty\ e^{-j\theta}$$

$$\underset{\varepsilon\to 0}{Lt}\ GH(\varepsilon e^{j\theta}) = \infty\ e^{-j(180+\theta)}$$

θ	$\underset{\varepsilon\to 0}{Lt}\ GH(\varepsilon e^{j\theta})$





Number of encirclements of point -1+j0, N = 0

P = 0

Z = P + N = 0 ; system is stable.

Sketch the Nyquist plot for the open-loop transfer function $G(s)H(s) = \frac{}{(s}$

hence determine the stability of the closed loop system by Nyquist criterion.

RELATIVE STABILITY USING NYQUIST CRITERION

As the Nyquist plot gets closer to -1+j0 point, the relative stability reduces and th

instability. In frequency domain, relative stability is characterised by gain marg

Gain margin (GM) is defined as the reciprocal of the magnitude of the open-

evaluated at the phase cross-over frequency ω_{pc}.

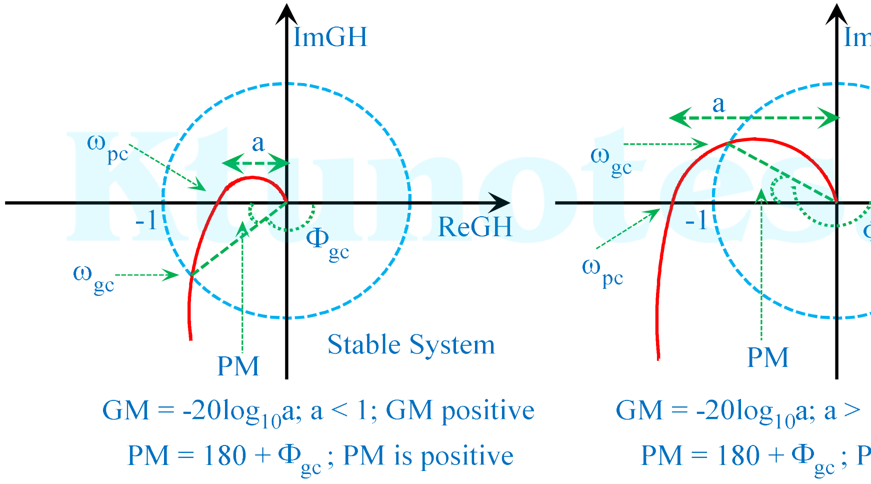
$$GM = \frac{1}{\left|G(j\omega_{pc})H(j\omega_{pc})\right|} = \frac{1}{a} \quad \text{where } \angle G(j\omega_{pc})H(j\omega_{pc}) = -180°$$

The frequency at which the phase angle of the open-loop transfer function is -1

cross-over frequency ω_{pc}.

$$PM = 180° + \angle G(j\omega_{gc})H(j\omega_{gc}) = 180° + \phi_{gc}$$

For a stable system, the phase margin is positive.



If $\omega_{gc} < \omega_{pc}$, system is stable.

If $\omega_{gc} = \omega_{pc}$, system is marginally stable.

If $\omega_{gc} > \omega_{pc}$, system is unstable.

ADVANTAGES OF NYQUIST PLOT

1	The Nyquist plot can be used for the study of stability of system with r functions.
2	The stability analysis of a closed loop system can be easily investiga Nyquist plot of the loop transfer function with reference to the -1+j0 made.

DISDVANTAGES OF NYQUIST PLOT

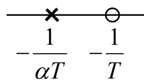
1	It is not so easy to carry out the design of the controller by referring to

compensative pole is also placed to reduce the noise.
The pole is located at a distance 3 to 10 times the value of the zero location.

Transfer function of lead compensator is

$$G_c(s) = \frac{s+z_c}{s+p_c} = \frac{s+1/T}{s+1/\alpha T};\ \alpha < 1$$

$$\alpha = \frac{z_c}{p_c} < 1$$



Angle contribution is always positive; hence the name lead.

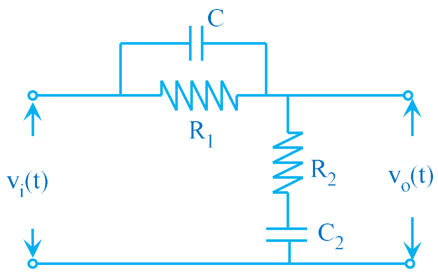
Eg:- $G_c(s) = \frac{s+1}{s+5}$;

$$\angle G_c(j\omega) = \tan^{-1}\omega - \tan^{-1} 0.2\omega$$

if ω=1, $\angle G_c(j\omega) = +33.7°$; here the angle is positive; hence LEAD.

Positive angle contributed by the lead compensator shifts the root locus toward
this results in improvement in the transient response.

A given system is stable but its transient response is unsatisfactory. Then, t
reshaped so that it is moved farther to the left, away from the imaginary axis. T
a lead compensator.



$$\frac{V_o(s)}{V_i(s)} = \frac{R_2}{R_2+Z_1} = \frac{R_2}{R_2 + \frac{R_1}{R_1Cs+1}} =$$

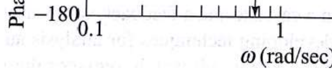
$$= \frac{R_1R_2C(s+\frac{1}{CR_1})}{R_1R_2C(s+\frac{R_1+R_2}{R_1R_2C})} = \frac{s+\frac{}{C}}{s+\frac{R_1+R}{R_2}}$$

where $T = CR_1$ & $\alpha = \frac{R_2}{R_1+R_2}$

The performance of a closed-loop system can be described in terms of the follo
specifications;

1) Phase Margin PM – indicative of relative stability
2) Gain cross over frequency ω_{gc} – indicative of speed of response (ω_{gc} is in

Since ω_{gc} decreases, speed of response decreases and since K_v decreases, steady state error increases- both transient & steady state requirements are not met.
Here, the desired specifications are not met by simple gain adjustment.



BODE PLOT OF LEAD COMPENSATOR

$$G_c(s) = \frac{s+\frac{1}{T}}{s+\frac{1}{\alpha T}} = \alpha \times \frac{1+sT}{1+s\alpha T}$$

Since α<1, lead compensator provides an attenuation of α; hence use suitable ar

$$G_c(s) = \frac{1+sT}{1+s\alpha T}$$

$$G_c(j\omega) = \frac{1+j\omega T}{1+j\omega\alpha T}$$

Corner frequencies are $\omega_{c1} = \frac{1}{T}$ and $\omega_{c2} = \frac{1}{\alpha T}$

ω rad/s	A in dB
$\omega \le \omega_{c1} = \frac{1}{T}$	A = 0
$\omega_{c2} = \frac{1}{\alpha T}$	$A = 20\log \omega T = 20\log \frac{1}{\alpha T} \times T = 20\log \frac{1}{\alpha}$
$\omega = \frac{10}{T}$	$A = 20\log \omega T - 20\log \alpha\omega T = 20\log \frac{1}{\alpha}$

$$\phi = \tan^{-1}\omega T - \tan^{-1}\alpha\omega T$$

At ω_m, $A = 20\log \omega T = 20\log \frac{1}{T\sqrt{\alpha}} \times T = 20\log \frac{1}{\sqrt{\alpha}} = 10\log \frac{1}{\alpha}$

$$\phi_m = \tan^{-1} \omega T - \tan^{-1} \alpha\omega T = \tan^{-1} \frac{1}{T\sqrt{\alpha}} \times T - \tan^{-1} \alpha \times \frac{1}{T\sqrt{\alpha}} \times T = \tan^{-1} \frac{1}{\sqrt{\alpha}} - \tan$$

$$\sin \phi_m = \frac{1-\alpha}{1+\alpha} \quad \text{or} \quad \alpha = \frac{1-\sin \phi_m}{1+\sin \phi_m}$$

	DESIGN OF LEAD COMPENSATOR - STEPS
1	The open loop gain K of the given system is determined to satisfy the state error.
2	After determining the value of K, draw Bode plot of uncompensated sys
3	Measure the gain cross-over frequency ω_{gc}, and phase margin PM c system.
4	Determine the amount of phase angle to be contributed by the lead formula given below: $\phi_m = \gamma_d - \gamma + \varepsilon$ where Φ_m = Maximum phase lead angle of lead compensator γ_d = desired phase margin γ = phase margin of uncompensated system ε = additional phase lead (margin of safety) to compensate for shi frequency due to compensation Choose an initial choice of ε as 5°. Note: If Φ_m is more than 60°, then realize the compensator as cascade of with each compensator contributing half of the required angle.
5	Determine the parameter α of the compensator using the following equa
6	Locate the frequency at which the uncompensated system has a gain of this frequency as the new gain cross-over frequency ω_{gc}'. Meas uncompensated system at ω_{gc}'. If the difference in PMs of G(jω) at ω_{g} go to next step. Otherwise choose a larger ε.
7	Set $\omega_m = \omega_{gc}'$ and compute parameter T of the compensator using the e

With K=15; $G(s)=\frac{\quad}{s(s+1)}$. Plot Bode plot.

$$G(j\omega)=\frac{15}{j\omega(1+j\omega)}$$

ω rad/s	A in dB
$\omega = 0.5$	$A = 20\log 15 - 20\log\omega = 29.5$dB
$\omega = \omega_{c1} = 1$	$A = 20\log 15 - 20\log\omega = 23.5$dB
$\omega = 10$	$A = 20\log 15 - 20\log\omega - 20\log\omega = -16.5$dB
$\omega = \omega_{gc}$, $A = 0$	$20\log 15 - 40\log\omega = 0$; $\omega_{gc} = 3.9$ rad/s

$$\phi = -90 - \tan^{-1}\omega$$

ω	1	2	5	10
Φ	-135	-153	-169	-174

$$\phi_{gc} = -90 - \tan^{-1}\omega_{gc} = -90 - \tan^{-1}3.9 = -166°$$ $$PM = 180 - 166 = 14°$$

$$\phi_m = \gamma_d - \gamma + \varepsilon = 45 - 14 + 5 = 36°$$

$$\alpha = \frac{1-\sin\phi_m}{1+\sin\phi_m} = \frac{1-\sin 36}{1+\sin 36} = 0.26$$

$$A = -10\log\frac{1}{\alpha} = -10\log\frac{1}{0.26} = -5.85$$

$$A = 20\log 15 - 40\log\omega_m = -5.85dB$$

$$\omega_m = \omega_{gc}' = 5.4rad/s$$

$$T = \frac{1}{\omega_m\sqrt{\alpha}} = \frac{1}{5.4\sqrt{0.26}} = 0.36$$

$$G_c(s) = \frac{1+sT}{1+s\alpha T} = \frac{1+0.36s}{1+0.094s}$$

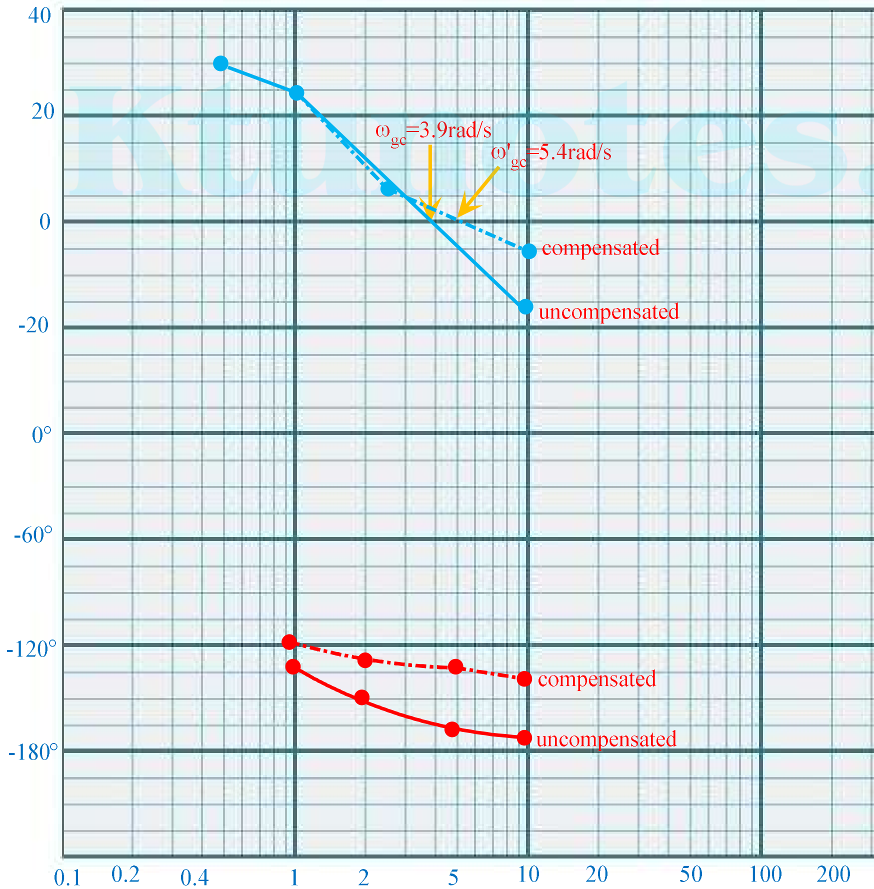
$$G_o(s) = \frac{1+sT}{1+s\alpha T}\times G(s) = \frac{1+0.36s}{1+0.094s}\times\frac{15}{s(1+s)}$$

$$G_o(s) = \frac{15(1+0.36s)}{s(1+0.094s)(1+s)}$$

ω	1	2	5	10
Φ	-121	-128	-133	-143

$PM = 180 - 133.6 = 46.4°$

All the requirements are satisfied.



$\omega = 0.5$	$A = 20\log 50 - 20\log\omega = 40dB$
$\omega = \omega_{c1} = 1$	$A = 20\log 50 - 20\log\omega = 33.98dB$
$\omega = \omega_{c1} = 5$	$A = 20\log 50 - 20\log\omega - 20\log\omega = 6dB$
$\omega = \omega_{c1} = 20$	$A = 20\log 50 - 20\log\omega - 20\log\omega - 20\log 0.2\omega =$ -3
$\omega = \omega_{gc}$, A = 0	$20\log 50 - 40\log\omega - 20\log 0.2\omega = 0;\ \omega_{gc} = 6.3$ r

$$\phi = \angle G(j\omega) = -90 - \tan^{-1}\omega - \tan^{-1}0.2\omega$$

ω	0.5	1	2	3	5	10	20	∞
ϕ	-122	-146	-175	-193	-214	-238	-253	-270

At $\omega = \omega_{pc}$, $\phi = -180$; $\phi = \angle G(j\omega) = -90 - \tan^{-1}\omega - \tan^{-1}0.2\omega = -180°$

At $\omega = \omega_{pc} = 2.24$ rad/s, $A = 20\log 50 - 40\log\omega - 20\log 0.2\omega = 26.94dB$

At $\omega = \omega_{gc} = 6.3$ rad/s, $\phi_{gc} = \angle G(j\omega) = -90 - \tan^{-1}\omega - \tan^{-1}0.2\omega = -22$

Phase Margin $\gamma = 180 + \phi_{gc} = 180 - 223 = -43°$

Since the phase margin is negative, system is unstable. A lead compensator is u stable and to have a phase margin of 20°.

$$\phi_m = \gamma_d - \gamma + \varepsilon = 20 + 43 + 5 = 68°$$

Since the lead angle required is greater than 60°, we have to realize the lead co of two lead compensators with each compensator providing half of the required

$$\phi_m = \frac{68}{2} = 34°$$

$$\alpha = \frac{1 - \sin\phi_m}{1 + \sin\phi_m} = \frac{1 - \sin 34}{1 + \sin 34} = 0.283$$

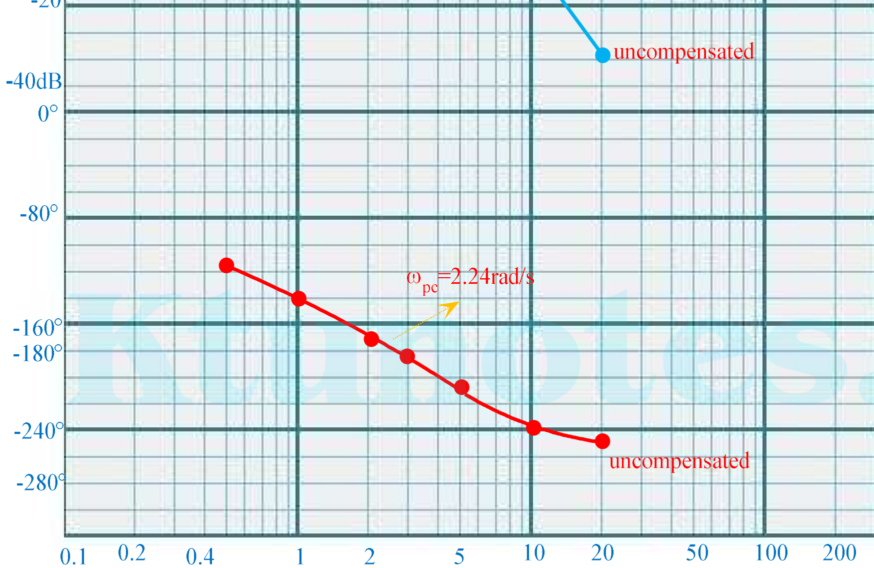
$$A = -10\log\frac{1}{\alpha} = -10\log\frac{1}{0.283} = -5.482$$

$$A = 20\log 50 - 40\log\omega_m - 20\log 0.2\omega_m = -5.483dB$$

$$\omega_m = \omega_{gc}' = 7.8 rad/s$$

$$T = \frac{1}{\omega_m\sqrt{\alpha}} = \frac{1}{7.8\sqrt{0.283}} = 0.24$$

$$G_c(s) = \frac{1 + sT}{1 + s\alpha T} = \frac{1 + 0.24s}{1 + 0.068s}$$

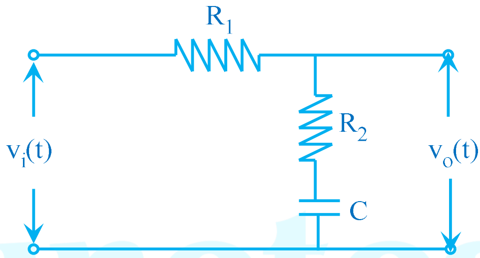


Design a lead compensator for the unity feedback system whose open loo

$G(s)=\dfrac{K}{s(s+2)^2}$. The following specifications are to be satisfied. i) $PM \geq 50°$

Angle contribution is always negative; hence the name lag.

$T \qquad \beta T$



$$G_c(s)=\frac{V_o(s)}{V_i(s)}=\frac{Z_2(s)}{Z_1(s)+Z_2(s)}=\frac{R_2+\dfrac{1}{Cs}}{R_1+R_2+\dfrac{1}{Cs}}=\frac{R_2C\left(s+\dfrac{1}{R_2C}\right)}{(R_1+R_2)C\left(s+\dfrac{1}{(R_1+R_2)C}\right)}$$

$$=\frac{\left(s+\dfrac{1}{R_2C}\right)}{\left(\dfrac{R_1+R_2}{R_2}\right)\left(s+\dfrac{1}{\left(\dfrac{R_1+R_2}{R_2}\right)R_2C}\right)}=\frac{\left(s+\dfrac{1}{T}\right)}{\beta\left(s+\dfrac{1}{\beta T}\right)}$$

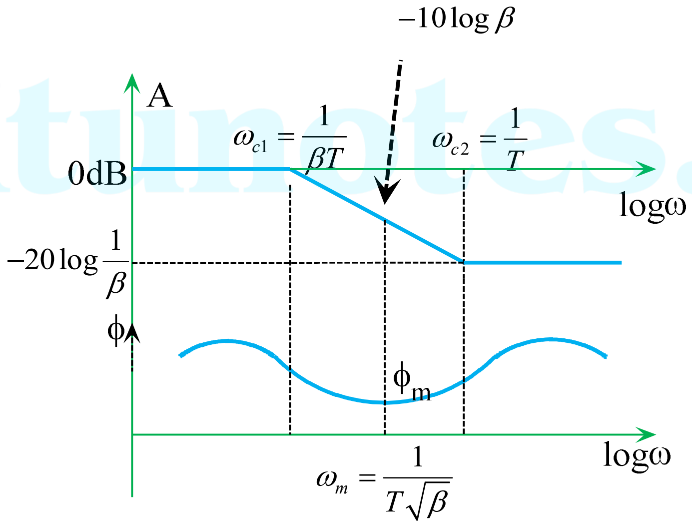
$$T=R_2C \qquad \text{and} \qquad \beta=\frac{R_1+R_2}{R_2}$$

$$G_c(s)=\frac{\left(s+\dfrac{1}{T}\right)}{\left(s+\dfrac{1}{\beta T}\right)}$$

Note: To nullify the attenuation $1/\beta$, it is amplified with gain β.

$\omega_{c2} = \frac{1}{T}$	$A = -20\log\omega\beta T = -20\log\beta$
$\omega = \frac{10}{T}$	A = – 20logωβT + 20logωT = – 20log10β + 20log1

$$\phi = \tan^{-1}\omega T - \tan^{-1}\beta\omega T$$



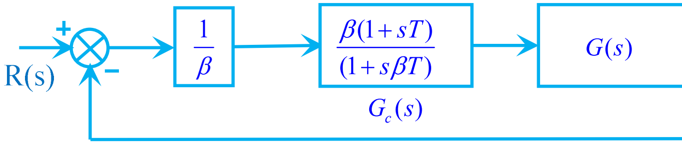
Φ is maximum at $\omega_m = \sqrt{\omega_{c1}\omega_{c1}} = \sqrt{\frac{1}{\beta T}\times\frac{1}{T}} = \frac{1}{T\sqrt{\beta}}$

At ω_m, $A = -20\log\omega\beta T = -20\log\frac{1}{T\sqrt{\beta}}\times\beta T = -20\log\sqrt{\beta} = -10\log\beta$

$$\phi_m = \tan^{-1}\omega T - \tan^{-1}\beta\omega T = \tan^{-1}\frac{1}{T\sqrt{\beta}}\times T - \tan^{-1}\beta\times\frac{1}{T\sqrt{\beta}}\times T = \tan^{-1}\frac{1}{\sqrt{\beta}} - \tan$$

$$\sin\phi_m = \frac{1-\beta}{1+\beta} \text{ or } \beta = \frac{1-\sin\phi_m}{1+\sin\phi_m}$$

DESIGN OF LAG COMPENSATOR - STEPS	
1	The open loop gain K of the given system is determined to satisfy the state error.
2	After determining the value of K, draw Bode plot of uncompensated sys

	$\beta = 10^{\ 20}$
7	Determine the transfer function of lag compensator. Zero of the lag compensator, $z_c = \frac{\omega_{gcn}}{10} = \frac{1}{T}$. (Place the zero of the com 1/10th of the new gain crossover frequency ω_{gcn}) Pole of the lag compensator, $p_c = \frac{1}{\beta T}$ Transfer function of lag compensator, $G_c(s) = \frac{1+sT}{1+s\beta T}$
8	Determine the open-loop transfer function of the compensated system 
9	Sketch the Bode plot of the compensated system and determine the actu actual phase margin satisfies the given specification, then the design is repeat the procedure by taking ε as 10°.

A unity feedback system has an open loop transfer function $G(s) = \frac{K}{s(1+2s)}$.
compensator so that phase margin is 40° and the steady state error for ramp inp
to 0.2.

$$e_{ss} = \frac{1}{K_v} = 0.2$$

$$K_v = 5$$

$$K_v = \underset{s \to 0}{Lt}\, sG(s) = \frac{K}{1} \geq 5$$

With K=5, $G(s) = \frac{5}{s(1+2s)}$. Plot Bode plot.

System requires a phase margin of 40˚, but the available phase margin is 18˚; h
should be employed to improve the phase margin.

$$\gamma_n = \gamma_d + \varepsilon = 40 + 5 = 45°$$

$$\phi_{gcn} = \gamma_n - 180° = 45 - 180 = -135°$$

$$\phi_{gcm} = -90 - \tan^{-1} 2\omega_{gcn} = -135°$$

Hence, new gain crossover frequency, $\omega_{gcn} = 0.5$ rad/sec

Zero of the compensator, $z_c = \dfrac{\omega_{gcn}}{10} = 0.05$

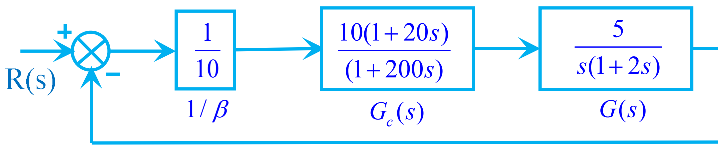
$$z_c = \frac{1}{T} = 0.05$$

dB magnitude at $\omega_{gcn} = 0.5$, $A_{gcn} = 20\log 5 - 20\log 0.5 = 20\text{dB}$

$$\beta = 10^{\frac{A_{gcn}}{20}} = 10$$

Pole of the lag compensator, $p_c = \dfrac{1}{\beta T} = 0.005$

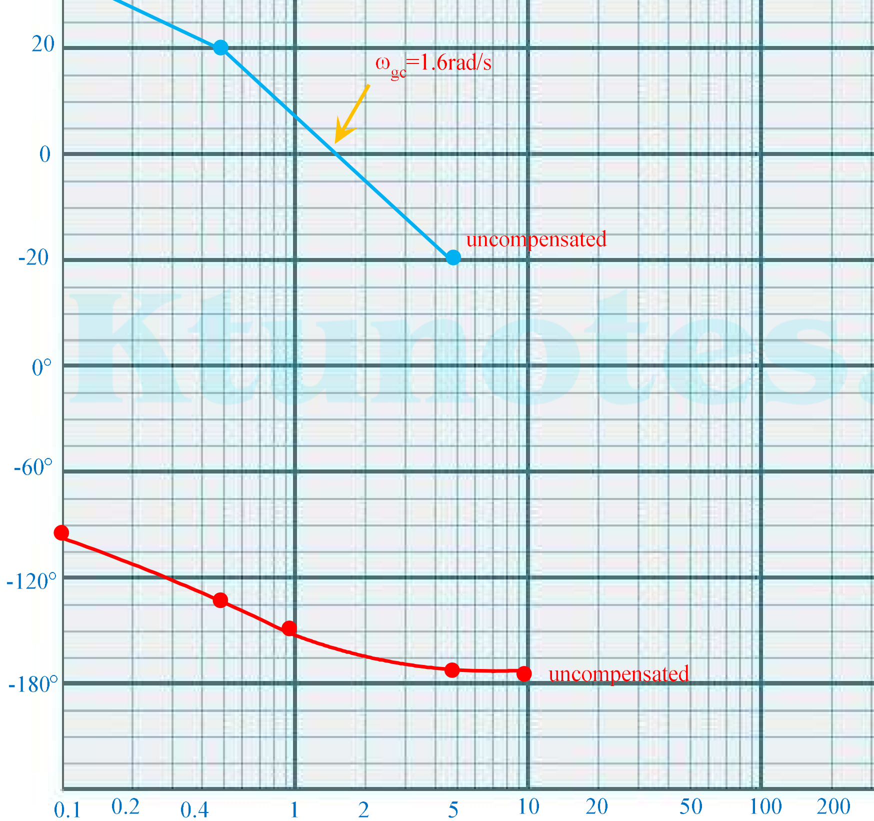
Transfer function of lag compensator, $G_c(s) = \dfrac{s + \frac{1}{T}}{s + \frac{1}{T}} = \beta \dfrac{1 + sT}{1 + s\beta T} = 10 \times \dfrac{1 + 20s}{1 + 200s}$



Overall transfer function, $G_o(s) = \dfrac{5(1 + 20s)}{s(1 + 200s)(1 + 2s)}$

Corner frequencies, $\omega_{c1} = 0.005$ rad/s; $\omega_{c2} = 0.05$ rad/s; $\omega_{c3} = 0.5$ rad/s;

ω rad/s	A in dB
$\omega = 0.001$	$A = 20\log 5 - 20\log\omega = 74\text{dB}$
$\omega = \omega_{c1} = 0.005$	$A = 20\log 5 - 20\log\omega = 60\text{dB}$



A unity feedback system has an open loop transfer function $G(s)=\dfrac{K}{s(s+4)(s+8}$

lag compensator so that phase margin is 33° and K_v = 30 sec^{-1}.

$$K_v = \underset{s\to 0}{Lt}\, sG(s) = \frac{K}{4\times 80} \geq 30$$

With K=9600, $G(s)=\dfrac{9600}{s(s+4)(s+80)}=\dfrac{30}{s(1+0.25s)(1+0.0125s)}$. Plot Bode plot.

Φ	-105	-129	-165	-193	-229	-254

$\phi_{gc} = -90 - \tan^{-1} 0.25\omega_{gc} - -\tan^{-1} 0.0125\omega_{gc} = -168°$

$\text{PM} = 180 - 168 = 12°$

System requires a phase margin of 30˚, but the available phase margin is 12˚; h
should be employed to improve the phase margin.

$\gamma_n = \gamma_d + \varepsilon = 33 + 5 = 38°$

$\phi_{gcn} = \gamma_n - 180° = 38 - 180 = -142°$

$\phi_{gcn} = -90 - \tan^{-1} 0.25\omega_{gcn} - \tan^{-1} 0.0125\omega_{gcn} = -142°$

Hence, new gain crossover frequency, $\omega_{gcn} = 4.6$ rad/sec

Zero of the compensator, $z_c = \dfrac{\omega_{gcn}}{10} = 0.46$

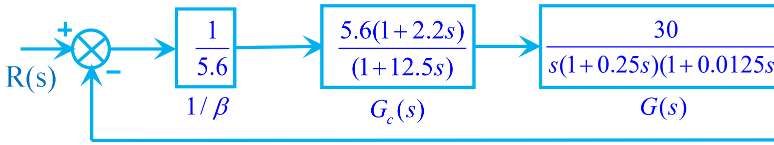
$z_c = \dfrac{1}{T} = 0.46$

dB magnitude at $\omega_{gcn} = 4.6$, A = 20log30 – 20logω – 20log0.25ω = 15dB

$\beta = 10^{\frac{A_{gcn}}{20}} = 5.6$

Pole of the lag compensator, $p_c = \dfrac{1}{\beta T} = 0.08$

Transfer function of lag compensator, $G_c(s) = \dfrac{s + \frac{1}{T}}{s + \frac{1}{T}} = \beta \dfrac{1 + sT}{1 + s\beta T} = 5.6 \times \dfrac{1 + 2.2s}{1 + 12.5s}$

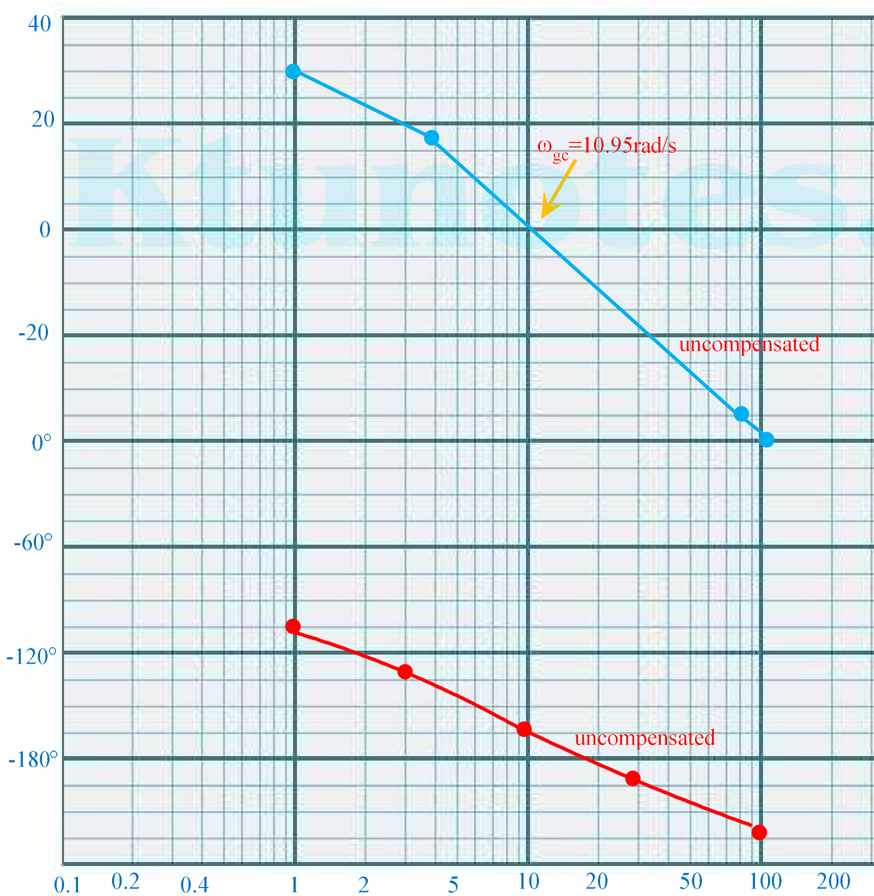


Overall transfer function, $G_o(s) = \dfrac{30(1 + 2.2s)}{s(1 + 12.5s)(1 + 0.25s)(1 + 0.0125s)}$

Corner frequencies, ω_{c1} = 0.08 rad/s; ω_{c2} = 0.46 rad/s; ω_{c3} = 4 rad/s; ω_{c2} = 80 ra

Actual phase margin of compensated system, $180° + \phi_{gc} = 180° - 147 = 33°$

Actual phase margin of the compensated system satisfies the requirement
acceptable.



A unity feedback system has an open loop transfer function $G(s) = \dfrac{K}{s(s+5)^2}$.
compensator so that phase margin ≥ 70° and K_v = 10 sec^{-1}.

$$R_1 + \frac{\ }{sC_1}$$

$$Z_2(s) = R_2 + \frac{1}{sC_2} = \frac{sR_2C_2 + 1}{sC_2}$$

$$G(s) = \frac{V_o(s)}{V_i(s)} = \frac{Z_2(s)}{Z_1(s) + Z_2(s)} = \frac{\dfrac{sR_2C_2+1}{sC_2}}{\dfrac{R_1}{sR_1C_1+1} + \dfrac{sR_2C_2+1}{sC_2}} = \frac{(sR_2C_2+1)(sR_1C_1}{sR_1C_2 + (sR_2C_2+1)(s}$$

$$= \frac{(sR_2C_2+1)(sR_1C_1+1)}{s^2R_1R_2C_1C_2 + s(R_1C_1 + R_2C_2 + R_1C_2) + 1}$$

Dividing both numerator and denominator by $R_1R_2C_1C_2$,

$$G(s) = \frac{\left(s + \dfrac{1}{R_1C_1}\right)\left(s + \dfrac{1}{R_2C_2}\right)}{s^2 + s\left(\dfrac{1}{R_1C_1} + \dfrac{1}{R_2C_2} + \dfrac{1}{R_2C_1}\right) + \dfrac{1}{R_1R_2C_1C_2}}$$

The transfer function of lag-lead compensator is given by

$$G_c(s) = \frac{\left(s + \dfrac{1}{T_1}\right)\left(s + \dfrac{1}{T_2}\right)}{\left(s + \dfrac{1}{\beta T_1}\right)\left(s + \dfrac{1}{\alpha T_2}\right)} = \frac{\left(s + \dfrac{1}{T_1}\right)\left(s + \dfrac{1}{T_2}\right)}{s^2 + s\left(\dfrac{1}{\beta T_1} + \dfrac{1}{\alpha T_2}\right) + \dfrac{1}{\alpha\beta T_1T_2}}$$

Comparing,

$$T_1 = R_1C_1 \qquad T_2 = R_2C_2 \qquad \frac{1}{\beta T_1} + \frac{1}{\alpha T_2} = \frac{1}{R_1C_1} + \frac{1}{R_2C_2} + \frac{1}{R_2C_1} \qquad \frac{1}{\alpha\beta T_1T_2} =$$

Hence,

$$\alpha\beta = 1$$

2	After determining the value of K, draw Bode plot of uncompensated sys
3	Determine the phase margin PM of the uncompensated system. If the satisfactory, then compensation is required.
4	Phase margin of compensated system, $\gamma_n = \gamma_d + \varepsilon$ where γ_d = desired phase margin ε = additional phase lag (margin of safety) to compensate for shift in gai due to compensation Choose an initial choice of ε as 5°.
5	$\phi_{gcn} = \gamma_n - 180°$ Determine the new gain cross-over frequency ω_{gcn} corresponding to uncompensated system.
6	Choose the gain cross over frequency of lag compensator $\omega_{gcl} > \omega_{gcn}$.
7	Determine the parameter β of the compensator. dB magnitude at $\omega_{gcl} = A_{gcl}$ $\beta = 10^{\frac{A_{gcl}}{20}}$
8	Determine the transfer function of lag compensator. Zero of the lag compensator, $z_c = \frac{\omega_{gcl}}{10} = \frac{1}{T_1}$. (Place the zero of the com 1/10th of the new gain crossover frequency ω_{gcl}) Pole of the lag compensator, $p_c = \frac{1}{\beta T_1}$ Transfer function of lag compensator, $G_{c1}(s) = \frac{s + \frac{1}{T_1}}{s + \frac{1}{\beta T_1}} = \beta \frac{1 + sT_1}{1 + s\beta T_1}$
9	Determine the transfer function of lead compensator. Take $\alpha = \frac{1}{\beta}$ Locate the frequency ω_m at which the uncompensated system has a gain

11	Sketch the Bode plot of the compensated system and determine the actual phase margin. If the actual phase margin satisfies the given specification, then the design is ... repeat the procedure by taking ε as 10°.

A unity feedback system has an open loop transfer function $G(s) = \frac{K}{s(s+3)(s+...}$ ... lag-lead compensator so that phase margin is 35° and K_v = 80 sec^{-1}.

$$K_v = \underset{s\to 0}{Lt}\, sG(s) = \frac{K}{3\times 6} \geq 80$$

K = 1440

With K=1440, $G(s) = \frac{1440}{s(s+3)(s+6)} = \frac{80}{s(1+0.33s)(1+0.167s)}$. Plot Bode plot.

$$G(j\omega) = \frac{80}{j\omega(1+j0.33\omega)(1+j0.167\omega)}$$

Corner frequencies, ω_{c1} = 3 rad/s; ω_{c2} = 6 rad/s

ω rad/s	A in dB
ω = 1	A = 20log80 – 20logω = 38dB
ω = ω_{c1} = 3	A = 20log80 – 20logω = 28.5dB
ω = ω_{c2} = 6	A = 20log80 – 20logω – 20log0.33ω = 16.6...
ω = 20	A = 20log80 – 20logω – 20log0.33ω – 20log0.167...
ω = ω_{gc} , A = 0	20log80 – 20logω – 20log0.33ω – 20log0.167... ω_{gc} = 11.3 rad/s

$$\phi = -90 - \tan^{-1} 0.33\omega - \tan^{-1} 0.167\omega$$

ω	1	3	10	30	100	300
Φ	-117	-161	-222	-253	-264	-268

$$\phi_{gc} = -90 - \tan^{-1} 0.33\omega_{gc} - -\tan^{-1} 0.167\omega_{gc} = -227°$$

PM $= 180 - 227 = -46°$

To determine the transfer function of lag section:-

New phase margin, $\gamma_n = \gamma_d + \varepsilon = 35 + 5 = 40°$

$\beta = 10^{\frac{\ }{20}} = 15$

Pole of the lag compensator, $p_c = \dfrac{1}{\beta T} = \dfrac{0.4}{15} = 0.027$; βT = 37

Transfer function of lag compensator, $G_1(s) = \dfrac{s + \frac{1}{T}}{s + \frac{1}{T}} = \beta \dfrac{1 + sT}{1 + s\beta T} = 15 \times \dfrac{1 + 2.5s}{1 + 37s}$

To determine the transfer function of lead section:-

$$\alpha = \frac{1}{\beta} = \frac{1}{15} = 0.067$$

$$A = -10\log\frac{1}{\alpha} = -10\log\frac{1}{0.067} = -11.7$$

$$A = 20\log 80 - 20\log\omega_m - 20\log 0.33\omega_m - 20\log 0.167\omega_m = -11.7dB$$

$$\omega_m = \omega_{gc}' = 17.7 rad/s$$

$$T_2 = \frac{1}{\omega_m\sqrt{\alpha}} = \frac{1}{17.7\sqrt{0.067}} = 0.22$$

$$\alpha T_2 = 0.067 \times 0.22 = 0.015$$

$$G_2(s) = \frac{1 + sT_2}{1 + s\alpha T_2} = \frac{1 + 0.22s}{1 + 0.015s}$$

Transfer function of the lag lead compensator is

$$G_c(s) = \frac{(1 + 2.5s)(1 + 0.22s)}{(1 + 37s)(1 + 0.015s)}$$

Transfer function of compensated system is

$$G_o(s) = \frac{(1 + 2.5s)(1 + 0.22s)}{(1 + 37s)(1 + 0.015s)} \times \frac{80}{s(1 + 0.33s)(1 + 0.167s)}$$

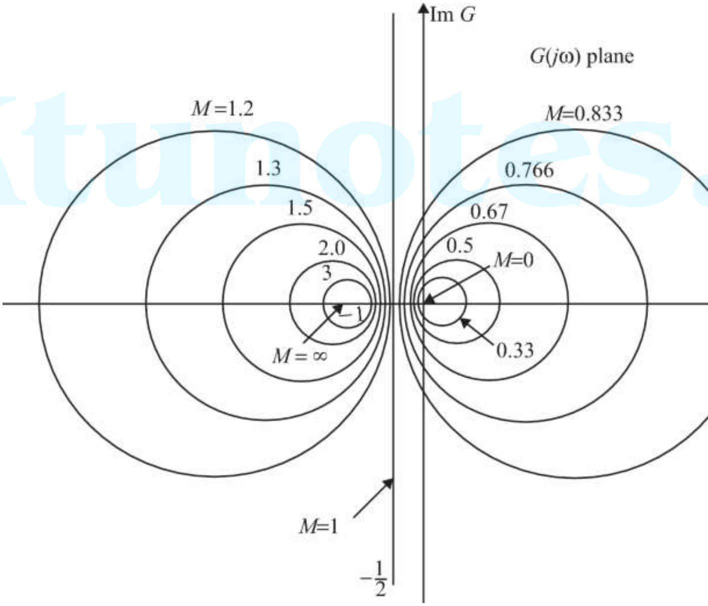
A unity feedback system has an open loop transfer function $G(s) = \dfrac{\ }{s(1+2s}$

suitable lag-lead compensator so that phase margin ≥ 55° and $K_v = 30\ sec^{-1}$.

$$\left[\quad 1-M^2\right] \qquad \left[1-M^2\right]$$

This represents the equation of a circle with the centre at $\left\{\frac{M^2}{1-M^2}, 0\right\}$ and havin

For a particular value of M, a circle is obtained. For various values of M, we These circles are known as constant magnitude loci or M circles.



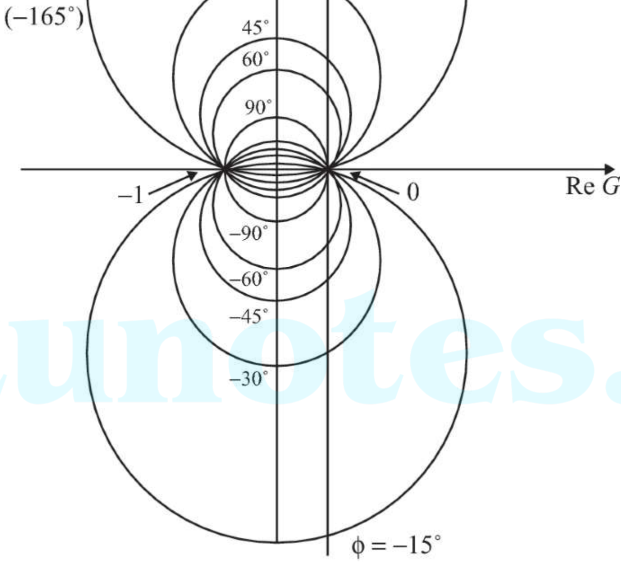
N CIRCLES (CONSTANT PHASE LOCI)

$$\angle M(j\omega) = \angle\frac{G(j\omega)}{1+G(j\omega)} = \angle\frac{X+jY}{1+X+jY}$$

$$\alpha = \tan^{-1}\frac{Y}{X} - \tan^{-1}\frac{Y}{1+X}$$

$$\tan\alpha = N = \tan\left[\tan^{-1}\frac{Y}{X} - \tan^{-1}\frac{Y}{1+X}\right]$$

Rearranging,



NICHOLS CHART

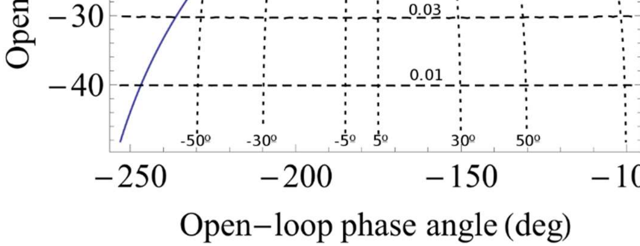
When we transform M and N circles to log magnitude and phase angle coordina known as Nichols Chart. The critical point (-1, j0) point is mapped to the Nic (0dB, -180˚). The Nichols chart contains curves of constant closed-loop magnit

The Nichols chart is symmetrical about the -180˚ axis. The M and N loci repeat f is symmetry at every 180˚ interval. The M loci are centered about the critical po

If we superimpose the gain-phase plot of an open-loop transfer function on Nic easily the closed-loop frequency response. Nichols chart gives the points of in phase plot of an open loop transfer function.

Nichols chart is used for the determination of the following: -

i) The complete closed-loop frequency response.
ii) Resonant peak of the closed loop system
iii) Resonant frequency of the closed loop system
iv) Bandwidth of the closed loop system
v) Phase margin



Open
−30
−40
0.03
0.01
-50º
-30º
-5º
5º
30º
50º
−250
−200
−150
−10
Open−loop phase angle (deg)

4	For the system $\frac{C(s)}{R(s)} = \frac{16}{s^2+4s+16}$, the damped frequency of oscillatio		
A	3.46	B	3.85
C	4.2	D	1.3
5	The type of the system having $G(s) = \frac{K(1+2s)}{s^2}$ and $H(s) = \frac{K(1+3s)}{s^2+2s+1}$		
A	Type 1	B	Type 2
C	Type 0	D	Type 3
6	The steady-state error of a unity feedback control system having open $G(s) = \frac{K}{s(s+2)}$ due to step input is		
A	Zero	B	K
C	1/K	D	∞
7	The transfer function of a plant is $G(s) = \frac{1}{s^2+0.2s+1}$. The plant settlin when it settles to within 2% of its final value is		
A	20 sec	B	40 sec
C	35 sec	D	45 sec
8	If $s^2+4s+16=0$ be the characteristic equation of a closed loop contro frequency in radians per second of the system is		
A	2	B	$2\sqrt{3}$
C	4	D	$2\sqrt{2}$
9	A system has forward path transfer function and feedback path tran $(s+5)$ respectively. The steady state error of the system for a unit step		
A	0	B	0.5
C	0.2	D	0.8
10	The impulse response of an RL series circuit is a		
A	Rising exponential function	B	Decaying exponen
C	Step function	D	Parabolic function
11	A synchro-transmitter receiver unit is a		

	The step error coefficient of a system $G(s) = \frac{\quad}{(s+1)(s+6)}$ with unity f		
A	1/6	B	∞
C	0	D	1
16	The transfer function of a system is $\frac{s+6}{Ks^2+s+6}$. The value of K for w will be 0.5 is		
A	1/3	B	3
C	1/6	D	6
17	A closed loop system having characteristic equation $s^2+2s+2=0$		
A	overdamped	B	Critically damped
C	Under damped	D	undamped
18	The transfer function of a phase lead compensator is $\frac{1+3Ts}{1+Ts}$. The ma provided by this compensator is		
A	90°	B	60°
C	45°	D	30°
19	The Nyquist plot of an open loop transfer function $G(j\omega)H(j\omega)$ of (-1, j0) point. The gain margin of the system is		
A	Less than zero	B	zero
C	Greater than zero	D	infinity
20	The open loop transfer functions with unity feedback are shown below (a) $G(s)=\frac{2}{(s+2)}$ (b) $G(s)=\frac{2}{s(s+2)}$ (c) $G(s)=\frac{2}{s^2(s+2)}$ (d) $G(s$		
A	1	B	2
C	3	D	4
21	For the transfer function $G(s)H(s)=\frac{1}{s(s+1)(s+0.5)}$, the phase cross		
A	0.5 rad/sec	B	0.707 rad/sec
C	1.732 rad/sec	D	2 rad/sec

26	The transfer function of a simple RC network functioning as a contr The condition for the RC network to act as a phase lead controller is		
A	p < z	B	p > 0
C	p = z	D	p > z
27	The transfer function of a tachometer is of the form		
A	Ks	B	$\frac{K}{s}$
C	$\frac{K}{1+s}$	D	$\frac{K}{s(1+s)}$
28	The AC motor used in servo application is a		
A	1-phase induction motor	B	2-phase induction
C	3-phase induction motor	D	Synchronous moto
29	A unity feedback system has the open loop transfer function $G(s) = —$ Nyquist plot of G(s) encircles the origin		
A	Never	B	Once
C	Twice	D	Thrice
30	A system is highly oscillatory if		
A	Gain margin is high	B	Gain margin is clo
C	Gain margin is 1 or phase margin is zero	D	Gain margin is hig
31	A system has high gain margin and phase margins. The system is		
A	Very stable	B	Sluggish
C	Very stable and sluggish	D	Oscillatory
32	In a minimum phase system,		
A	All poles lie in the left half of s-plane	B	All zeros lie in the
C	All poles & zeros lie in the left half of s-plane	D	All except one pol half of s-plane
33	The addition of a pole to the open loop transfer function pulls the root		
A	To the right	B	To the left and ma stable

A	Lead compensator	B	Lag compensator
C	Either A or B	D	Neither A nor B
38	The transfer function $G_c(s) = \dfrac{3(s+0.05)}{s+0.005}$ can be for		
A	Lead compensator	B	Lag compensator
C	Either A or B	D	Neither A nor B
39	In Bode plot, an octave is a frequency band from		
A	ω_1 to $10\omega_1$	B	ω_1 to $8\omega_1$
C	ω_1 to $4\omega_1$	D	ω_1 to $2\omega_1$
40	For the transport lag $G(j\omega) = e^{-j\omega T}$, the magnitude is always equal to		
A	0	B	1
C	2	D	3
41	For the transport lag $G(j\omega) = e^{-j\omega T}$, the polar plot is a		
A	Semi circle	B	Circle
C	Unit circle	D	None of these
42	A lead compensator is basically a		
A	High pass filter	B	Low pass filter
C	Band stop filter	D	None of these
43	The primary function of a lag compensator is to provide sufficient		
A	Gain margin	B	Phase margin
C	Both A and B	D	Either A or B
44	A lag compensator is basically a		
A	High pass filter	B	Low pass filter
C	Band stop filter	D	None of these
45	In second order system with a unit step input, the speed of response is		
A	Overdamped	B	Undamped
C	Critically damped	D	Under damped
46	In a second order system, the time constant of exponential envelopes d		
A	Only on damping factor	B	Only on natural fre
C	Both on damping factor & natural frequency	D	Neither on dampin frequency

50	The system described by the equation $s^4 + 2s^3 + 3s^2 + 6s + K = 0$ accor criteria, is		
A	Unstable for all values of K	B	Stable if $K \geq 0$
C	Stable if $K < 0$	D	Stable for all value
51	The root locus of a system has 4 separate loci. The system can have		
A	4 poles & 4 zeros	B	4 poles OR 4 zeros
C	6 poles & 2 zeros	D	2 poles & 2 zeros
52	The gain cross over frequency and bandwidth of a control syst respectively. A phase lag network is employed for compensating the sy over frequency and bandwidth of the compensated system are ω_{cc} and		
A	$\omega_{cc} < \omega_{cu}$ & $\omega_{bc} < \omega_{bu}$	B	$\omega_{cc} < \omega_{cu}$ & $\omega_{bc} >$
C	$\omega_{cc} > \omega_{cu}$ & $\omega_{bc} < \omega_{bu}$	D	$\omega_{cc} > \omega_{cu}$ & $\omega_{bc} >$
53	Laplace transform of the output response of a linear system is the sys when the input is		
A	A step signal	B	A ramp signal
C	An impulse signal	D	A sinusoidal signa
54	A system has the following transfer function $G(s) = \frac{100(s+5)(s+}{s^4(s+10)(s^2+3}$ order of the system are respectively		
A	4 and 9	B	4 and 7
C	5 and 7	D	7 and 5
55	The open loop transfer function of a unity feedback system is $G(s$ margin of the system will be		
A	20dB	B	40dB
C	60dB	D	80dB
56	A system has single pole at origin. Its impulse response will be		
A	constant	B	ramp
C	Decaying exponential	D	oscillatory
57	Consider the following techniques: 1. Bode plot		

60	Which one of the following statements is correct? Nichols chart is useful for detailed study and analysis of		
A	Closed loop frequency response	B	Open loop frequen
C	Closed loop and open loop frequency response	D	None of the above

Answers:

1	B
2	B
3	B
4	A
5	B
6	A
7	B
8	C
9	C
10	B
11	D
12	C
13	B
14	C
15	A
16	C
17	C
18	D
19	A
20	C

21	B
22	B
23	D
24	B
25	C
26	D
27	A
28	B
29	B
30	C
31	C
32	C
33	A
34	A
35	A
36	C
37	A
38	B
39	D
40	B

1 Give a comparison between open loop and closed loop control sy
examples.

2 Derive the dependence of φ_m and α of a lead compensator and
restrictions on the selection of α?

3 For a closed loop system with $G(s) = \frac{1}{s(s+5)}$; and H(s) = 0.05, calcu
error constants.

4 Check the stability of the system given by the chara
$G(s) = s^5 + 2s^4 + 4s^3 + 8s^2 + 16s + 32$; using Routh criterion.

5 With suitable sketches explain how the addition of poles to the
function affect the root locus plots.

6 Explain Ziegler – Nichol's PID tuning rules.

7 Explain the features of non-minimum phase systems with a suitable e

8 How do you determine the gain margin of a system, with the help of

9 State and explain Nyquist stability criterion.

10 Discuss the procedure for Lag compensator design using Root locus

PART B

Answer any one full question from each module. Each question c

Module 1

11 a) Derive the transfer function of an Armature controlled dc servo
of time constants on the system performance.

b) Compare the effect of H(s) on the pole-zero plot of the cl
$G(s) = \frac{s+3}{(s^2+3s+2)}$ with: i) derivative feed back H(s)= s;
H(s)=1/s.

12 a) Why compensation is necessary in feedback control system?
be considered for choosing the feedback compensation?

b) With relevant characteristics explain the operation of the foll
i) Synchro error detector, ii) Tachogenerator.

b) Using Routh criterion determine the value of K for which the
loop system with $G(s) = \frac{K}{s(s^2 + 20s + 8)}$ is stable. .

Module 3

15 a) Design a lag lead compensator with open loop transfer functi

to satisfy the following specifications (i) damping ratio of the
poles is 0.5 (ii) Undamped natural frequency of the dominant
5 rad/sec iii) Velocity error constant $K_v = 80$.

b) Compare between PI and PD controllers.

16 a) Sketch root locus for a system with $G(s)H(s) = \frac{K(s+1)}{s(s+4)}$

range of K for the system stability. .

b) With help of suitable sketches, explain how does Angle and
Root locus method help in control system design.

Module 4

17 a) The open-loop transfer function of a unity fe
$G(s)=\frac{K}{s(0.5s+1)(0.04s+1)}$.Use asymptotic approach to plot
determine the value of K for a gain margin of 10 dB.

b) Compare between the polar plots for $G(s)H(s) = \frac{K}{(s+4)}$ and G

18 a) Draw the polar plot of an open loop transfer function G(s)= (s-

on the phase margin and gain margin.

b) Explain the detrimental effects of transportation lag, using Bo

Module 5

19 a) Draw Nyquist plot for the system whose open loop transfer fu
$\frac{K}{s(s+2)(s+10)}$. Determine the range of K for which the closed
(9)

b) Write a short note on Nichols chart.